

8.1 Integration by Parts.

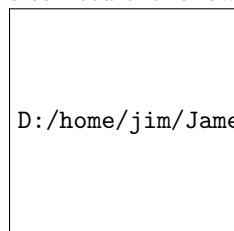
- Definite and indefinite integral of functions with $\ln x$
 - **20, 33, 51; ex.** $\int \ln 2x \, dx, \int_1^e \ln 2x \, dx$. Note that these examples, like problem 51, require substitution.
- You will still be expected to be able to use integration by parts
 - **ex.** $\int x^2 \ln x \, dx$.
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8.2 Trigonometric Integrals.

- Integrals of $\sin x$ or $\cos x$ with even or odd powers.
 - Even powers can be solved with the half-angle formula.
 - 9, 10; ex.** $\int \cos^4 2x \, dx$.
 - Odd powers can be solved by 1) splitting off a single power, e.g. $\sin^5 x = \sin^4 x \sin x$, 2) the Pythagorean identity, 3) substitution.
 - 11, 12; ex.** $\int \sin^5 x \, dx$.
- Integrals of products of $\sin x$ and $\cos x$ with even and/or odd power.
 - Problems with *only* even powers can be solved with the half-angle formula.
 - 13, 17; ex.** $\int \sin^4 x \cos^2 x \, dx$.
 - Problems with at *least one* odd power can be solved by 1) splitting off a single power, e.g. $\sin^5 x = \sin^4 x \sin x$, 2) the Pythagorean identity, 3) substitution.
 - 15, 14; ex.** $\int \cos^3 x \sin^2 x \, dx$.

8.3 Trigonometric Substitutions

Integrals involving $a^2 - x^2$. The important substitutions are: $x = a \sin \theta$ and $\sqrt{a^2 - x^2} = a \cos \theta$. You may also need the following reference triangle:



- Definite integrals involving $a^2 - x^2$.
 - **7-10; ex.** $\int_0^3 \sqrt{36 - x^2} \, dx$.
- REMEMBER that the limits of integration need to be changed in these problems: $x = a \sin \theta \Rightarrow \sin^{-1}\left(\frac{x}{a}\right) =$ new endpoint (above $a = 3$, so $\frac{3}{2} = 3 \sin \theta \Rightarrow \sin^{-1}\left(\frac{3/2}{3}\right) = \frac{\pi}{6}$).
- Indefinite integrals involving $a^2 - x^2$.
 - **11-17; ex.** $\int \frac{dx}{(9 - x^2)^{1/2}}$.
- REMEMBER that the final result must be in terms of x (there should be no θ 's left).